

J-DRUMS

The electronic drum MIDI module built from scratch with
Arduino Mega 2560 — powerful, configurable, and open-source.

Arduino Mega 2560

MIT License

16 Analog Pads

MIDI USB + Serial

Smart Hi-Hat

ABOUT THE PROJECT

A complete and customizable MIDI module

J-DRUMS V12 is an electronic drum MIDI controller built for the Arduino Mega 2560. It turns piezos and sensors into professional MIDI signals — with an on-screen LCD menu, individual per-pad adjustment, a smart hi-hat via TCRT5000, and compatibility with all the leading VST software on the market.

KEY FEATURES

Everything you need in the kit



16 Analog Pads (A0-A15)

Each pad has an individually configurable MIDI note, threshold, gain, velocity curve, and crosstalk cancellation.



Smart Hi-Hat (TCRT5000)

Infrared optical sensor controls hi-hat opening in real time — auto-calibration in 8 seconds with A+B. Supports **two modes via InvSensor**: *Normal* (sensor manually assembled with a resistor) and *Inverted* (ready-made module with LM393 — e.g. TCRT5000 Arduino LM393 5V sensor). Calibration works the same in both modes.



3 Factory Presets with Ready-Made MIDI Maps

Addictive, SupEzd1, SupEzd2 — mapping switches automatically when you select the preset.



Dual MIDI Output

USB and Serial TX1 work simultaneously — connect to your PC and a sound module at the same time.



Digital Pad Expansion

7 built-in digital pads (pins D33–D45) + expansion via Pro Micro (4 pads) or Leonardo (6 pads) over SysEx.



3 Preset Slots in EEPROM

Save and restore complete kit configurations — notes, thresholds, chokes, and everything else with one click.



XCancel CrossTalk

Eliminates ghost notes from mechanical vibration — configurable Source→Target per pad without losing dynamics.



Quick Pad Adjust

Hold the encoder for 3s, hit the pad, and the menu jumps straight to that pad's settings — no menu navigation needed.



Real-Time Monitor

MIDI Monitor shows port, note, and velocity for each hit on the LCD. Hi-Hat Monitor displays a 16-position visual bar.



Rimshot, Tri-Zone, and Dual Pad — Hi-Hat, Ride, and Crash with 1 piezo each

A complete multi-zone system using digital membranes (ZoneDual/Tri-Zone). A single piezo per cymbal detects body, edge, and bell separately.

HI-HAT — TRI-ZONE

Piezo on **A1** + 2 digital membranes:

- Edge/Rim → free piezo (note 8)
- Body → Dig 3 / P41 (note 7)
- Bell → Dig 4 / P39 (note 9)

Pedal opening via HHC (A0) works correctly across all 3 zones.

RIDE — TRI-ZONE + CHOKE

Piezo on **A8** + 2 digital membranes:

- Bow → free piezo (note 51)
- Bell → Dig 1 / P45 (note 53)
- Ride Crash → Dig 2 / P43 (note 59)
- Ride Choke → Dig 2 / P43 (note 63) — holding down the Ride Crash membrane

Real piezo velocity is inherited across all 4 zones.

🔊 CRASH — DUAL ZONE (×2)

Crash 1 (A10) + Dig 5 / P37:

- Edge → free piezo (note 49)
- **Body** → membrane (note 27)

Crash 2 (A11) + Dig 6 / P35:

- Edge → free piezo (note 57)
- **Body** → membrane (note 31)

🥁 RIMSHOT — 2 MODES

Full rimshot:

Rim (A3) + Snare (A2) together → automatically triggers the rimshot note (40).

Force rimshot:

Snare (A2) at velocity 127 → automatically triggers rimshot, no rim hit needed.

Each mode is enabled independently in the menu.

⚠️ RESERVED PORTS — A0, A1, A2, A3, A8, A10, A11

These ports have notes reserved by the system and **should not be changed by the user**:

A0 = HHC — Hi-Hat Controller (note managed by the Hi-Hat system)

A1 = Hi-Hat (note managed by the Hi-Hat system)

A2 = Snare (note managed by the Rimshot system)

A3 = Snare Rim (note managed by the Rimshot system)

A8 = Ride Tip (note managed by the Ride's Tri-Zone system)

A10 = CRASH 1 (note managed by Crash 1's Dual-Zone Body system)

A11 = CRASH 2 (note managed by Crash 2's Dual-Zone Body system)

Changing these notes can break hi-hat and rimshot operation.

⌘ **80ms time latch** — even if the stick touches the membrane for just 3-10ms, the system keeps the zone active long enough for the piezo to process the hit. Works with any homemade membrane: copper tape, EVA foam, sheet metal.



Cymbal Choke

4 chokes on pins D47-D53 — triggered by direct metal contact, no extra components needed.



InvSensor — Hi-Hat Sensor Mode

Supports two sensor types for the Hi-Hat Controller (A0): **Normal** — manually assembled sensor with transistor and resistor (TCRT5000 + R10kΩ); **Inverted** — ready-made module with integrated LM393 (e.g. TCRT5000 Arduino LM393 5V Infrared Sensor). Auto-calibration works the same in both modes. Configured under: Pad A0 → InvSensor. Saved to EEPROM.



Custom Velocity Curves

7 curves available per pad: Linear, Soft1/2/3, Hard1/2, and Fixed — adapt the response to your playing style.



Fi-TipBell — Ride Tip×Bell Dispute Filter

An alternative to membrane Tri-Zone for the Ride: two separate piezos — tip on A8, bell on a second small cymbal on A9, isolated by just a nut between them. When both trigger within 12ms of each other, the firmware compares hit energy (velocity ÷ Gain) and sends only the note from whichever piezo was actually hit, with an adjustable bonus (0-30) to compensate for the bell's mechanically weaker signal. Only active on A8, and only while the Ride's Tri-Zone is Off.



Noise vs Real Hit Filter

A piezo never reads total silence — it picks up vibration from fans, TVs, switching power supplies, and the mains itself (60Hz). J-DRUMS detects the sharp variation of a real hit and ignores slow oscillations, blocking false triggers without touching real dynamics.

V12

Current firmware version — Arduino Mega 2560

16 1 4 3 4 16

ANALOG PADSDIGITAL PADCHOKESPRESET SLOTSMIDI MAPSXCANCEL PAIRS

54+

Tested and approved by a group of over 54 members, including professional musicians, drummers, and percussionists with years of experience, who test J-DRUMS on acoustic kits converted to electronic and use it in real practice.

The code reached such a high level that many entry-level modules on the market, with more hardware resources, deliver less performance and fewer features than J-DRUMS. Even running on an Arduino Mega 2560, the result achieved gets close to the level of top-tier professional modules, such as the TD-27 and similar.

MODULE NAVIGATION

Full menu on the LCD display

 PADS A0-A15

 Hi-Hat Controller

 Quick Pad Adjust

 Settings

 Display Backlight

 Dual Pad

 Save Presets

 Menu Sound

 MIDI Output

 Auxiliary Pads

 Invert Buttons

 MIDI Monitor

 Hi-Hat Monitor

 InvSensor (A0)

 XCancel CrossTalk

 Reset Module

 Trizone Notes

 Choke Notes

⚙ SETTINGS

All submenus inside Settings

 Summary of every submenu available inside Settings — in firmware order.

 **Display Backlight**

Turns off the LCD after 30s of inactivity, or keeps it always on

 **Dual Pad**

Enables/disables dual notes from a single piezo (101→102, 103→104)

Save Presets

Saves or restores the configuration in one of 3 EEPROM slots. Also includes 3 fixed factory presets (Addctive, SupEzd1, SupEzd2) that can never be erased.

Menu Sound

Enables or disables the audio feedback beeps during navigation

MIDI Output

Enables MIDI output via USB and/or TX1 — both can be used simultaneously

Auxiliary Pads

Enables pad expansion via Pro Micro (4 pads) or Leonardo (6 pads) over SysEx

Invert Buttons

Swaps the function of Buttons A and B between pins 6 and 7 — saved to EEPROM

Trizone Notes

Sets the MIDI note for the 7 digital pads on pins D45,43,41,39,37,35,33

Choke Notes

Sets the MIDI note for the 4 cymbal chokes on pins D53,51,47,49

Reset Module

Restores every value to factory default — double confirmation with a 10-second timeout each

MIDI Monitor

Monitors in real time the analog port (A0–A15), the MIDI note, and the velocity of every hit — MIDI keeps working normally during monitoring

Hi-Hat Monitor

Monitors hi-hat opening via the TCRT5000 with a 16-position visual bar — 1 block = maximum opening, 16 blocks = fully closed. Line 0: A<118> Hi F<001> with calibration adjustable from the menu.

CC MIDI Preset

Saves or restores 3 independent presets with the full CC MIDI Controller map — the 18 Mixes (button+knob) and the 3 extra Knobs (4 banks). Unlike the Save Presets menu (which stores the pads' general configuration), this one stores only the CC control values — ideal for quickly switching between different mixes in your VST/DAW.

TECHNICAL SPECIFICATIONS

Hardware and circuit

Microcontroller

Arduino Mega 2560

Main

Analog pads

27mm or 35mm ceramic piezo — 1MΩ resistor + 5V1 Zener

Hi-Hat sensor	TCRT5000 — port A0 — reading 0 to 1023 — InvSensor: Normal (manual assembly with resistor) or Inverted (module with LM393 — e.g. TCRT5000 Arduino LM393 5V)	Auto-calibrates
Digital/aux pads	Piezos on pins D33, D35, D37, D39, D41, D43, D45 — INPUT_PULLUP — 15ms debounce	
Cymbal chokes	Pins D47, D49, D51, D53 — direct metal contact, no external components	
MIDI Out	USB (native) + Serial TX1 — simultaneous	
Display	16×2 LCD with automatic backlight-off after 30s of inactivity	
Storage	Internal EEPROM — 3 presets + settings + all pad notes	
Expansion	Pro Micro (+4 pads) or Leonardo (+6 pads) via SysEx	
Indicator LED	Pin 22 + 220Ω resistor — blinks on every MIDI note generated	
License	MIT License — open-source, free to use and modify	Open Source

SOURCE CODE & MANUAL

Download the
complete project for free



J-DRUMS V12 — Complete Package

The download includes the **full source code** for the Arduino Mega 2560 and the **interactive Illustrated HTML Manual** — with every section, circuit diagrams, explained parameters, and configuration examples.

↓ J-DRUMS V12 Code Only

Arduino firmware, ready to upload

🔊 MIDI System — How it works

🔊 Sends MIDI to PC / DAW

J-DRUMS sends real-time MIDI signals to any drum software — Ableton, Reaper, FL Studio, **Pro Tools, ASIO / ASIO4ALL**, and all compatible VSTs (Superior Drummer, EZdrummer, Addictive Drums).

🔊 Properly implemented Note OFF

The firmware sends **Note ON** when it detects the hit and **Note OFF** right after — exactly as the MIDI standard requires. Cymbals and notes never get "stuck" in the VST.

🔊 Editable device ID

The **module's MIDI ID is editable** directly in the code — you can customize the name that appears in the operating system and DAW when the Arduino is connected via USB, tailoring it to your setup.



Arduino Source Code

Complete J-DRUMS V12 firmware for Arduino Mega 2560. Ready to upload with the Arduino IDE.



Interactive HTML Manual

Complete illustrated manual in HTML — opens in any browser. Includes circuit diagrams, every parameter explained, examples, and password-protected editable notes.



Circuit Diagrams

Wiring schematics for the analog pads, TCRT5000 sensor, digital pads, chokes, buttons, encoder, and LEDs.



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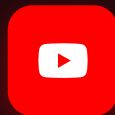
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Desenvolvido por **Joanaldo Jhon Leonez de Melo**

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